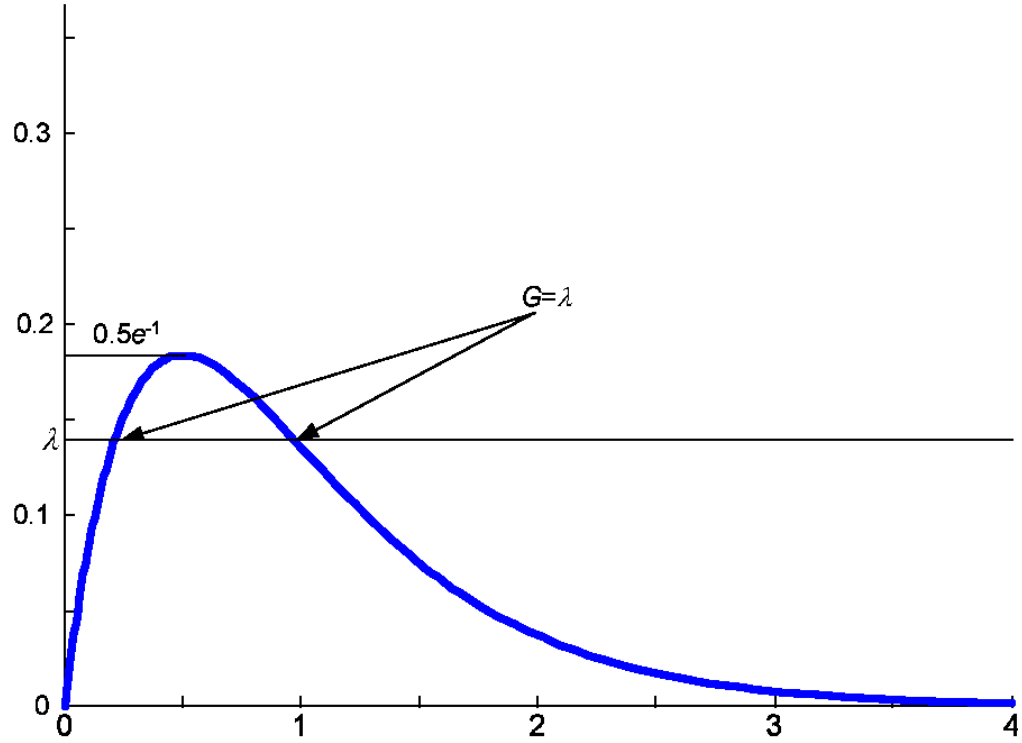
# Multiaccess and ALOHA

2006-11-14

Pure ALOHA, packets collide with any other packet transmitted within one time unit of their transmission. The diagram shows a packet sent at $t$ and packets sent at $t+1$ and $t-1$. Any packet sent after $t-1$ and before $t+1$ will collide with the packet sent at $t$.




Throughput for pure ALOHA as a function of $G$.



# Multiaccess Systems

- Finally we briefly review multiaccess systems.
- These are systems where there is one communication channel and multiple stations (or nodes) making use of it.
- Examples of multiaccess systems include: Satellite systems; packet radio networks; a multitapped bus.

# LANs

- A very common example of multi-access channels is the local area network.
- In a local area network transmission rates are around 10Mbps to 1Gbps.
- Transmission distances range from a few meters to several kilometres.

# LANs (continued)

- The required error rates are low: typically from $10^{-8}$ to $10^{-11}$, that's about 1 error every Gigabyte.
- Typically we are interested in the throughput of the network — the amount of data or messages passing through the network each second.
- If the number of overhead bits (associated with network administration *etc.*) is low then this is approximately equal to the utilization factor $\rho$.

# Channel Multiplexing

- Until the introduction of the ALOHA network the standard approach to splitting a channel was through multiplexing.
- We will now briefly review this approach to multiplexing using Example 3.9 from Bertsekas and Gallagher.
- Assume we have $m$ statistically independent Poisson streams.

# Channel Multiplexing (continued)

- Each has arrival rate $\frac{\lambda}{m}$.
- The packets to be transmitted are assumed to have an exponentially distributed length.
- For a given bandwidth this leads to a service time distribution for the whole transmission line which is an exponential with a rate parameter $\mu$.

# Statistical Multiplexing

- System 1: merge the Poisson streams together, through statistical multiplexing.
- The superposition property of the Poisson distribution tells us that they become a Poisson stream with rate $\lambda$.
- If the transmission times are exponentially distributed the average delay, from the $M/M/1$ queue, will be

$$\bar{T} = \frac{1}{\mu - \lambda}.$$

# Time or Frequency Division Multiplexing

- System 2: provide each communication channel with a separate 'circuit'.
- The transmission line needs to be split into $m$ separate channels.

# Time or Frequency Division Multiplexing (continued)

- Each of these separate channels will now have service time distributions with a reduced rated $\frac{\mu}{m}$.
- Delay in this system will now be

$$\bar{T} = \frac{1}{\frac{\mu}{m} - \frac{\lambda}{m}} = \frac{m}{\mu - \lambda}$$

which is $m$ times larger than that for statistical multiplexing.

# Why does this happen?

- The reduction in delay arises from the fact that when only one of the $m$ channels is transmitting with statistical multiplexing it can use the entire bandwidth of the channel but with time or frequency division multiplexing $\frac{m-1}{m}$ of the bandwidth is wasted.

# Multiaccess Systems

- In multiaccess systems splitting the channel using time division multiplexing is known as time division multiple access (TDMA).
- Similarly splitting the channel using frequency division multiplexing is known as frequency division multiple access (FDMA).

# Multiaccess Systems (continued)

- We have already shown that statistical multiplexing can make more efficient use of bandwidth than time or frequency based multiplexing.
- However in a multiaccess system it can be difficult to implement in the manner discussed above.

# Multiplexing for Multiaccess

- We assumed that the all the transmissions could be queued in a single $M/M/1$ queue.
- In a local area network the transmitting stations (or nodes) are distributed across different portions of the network as are the receiving nodes.

# Multiplexing for Multiaccess (continued)

- Forming a single queue is not possible without some form of centralised control.
- TDMA and FDMA can be implemented by by dedicating one channel for each station to transmit on, but allowing each station to receive on any channel.
- This allows for an analysis of the type we described above. However, it is not clear how to perform statistical multiplexing in these circumstances.

# Random Access

- When we have stations distributed across a single communication channel would be to have a centralised computer attempting to control which computer is able to access our transmission lines at what times.
- However, we may wish communication to occur without centralised control. In this case we may choose a *random access* solution.

# Random Access (continued)

- We will consider random access networks where there is one communication channel on which all the nodes communicate. [1]
- Each station sends a packet immediately whenever it has a packet to send. This means that packets can collide and may have to be resent.

[1] It could be the case that only a subset of the nodes have access to the channel.

# Pure ALOHA Networks

- ALOHA network was developed at the University of Hawaii.
- Packet radio network which communicated between different stations on the Hawaiian islands.
- For the ALOHA network
- all stations attempt to communicate on the same channel.

# Pure ALOHA Networks (continued)

- When a station attempts to transmit, it detects if its transmission has interfered with an existing transmission: this is known as a *collision*.
- If a collision occurs the station becomes *backlogged*and attempts to transmit at a later time.
- If both stations attempted to retransmit after the same period of time they would simply collide again.
- To avoid this problem the stations normally wait for a random period of time before retransmitting.

# ALOHA




pureALOHA

# ALOHA (continued)

- For this network we assume that each station can tell whether the transmitted packets were correctly received.
- Furthermore, we assume that each packet requires 1 'time unit' for transmission. This means that if a packet of interest is sent at time $t$, then any other packet sent between time $t-1$ and time $t+1$ will collide with our packet.

# Simple Analysis

- The arrival rate for packets at all stations is a Poisson process with rate $\lambda$.
- We assume that the number of stations is very large (infinite) so that an individual station's packet rate is an insignificant proportion of $\lambda$, *i.e.*

$$\frac{\lambda_n}{\lambda} = \lim_{\delta \to 0} \delta = 0.$$

- We denote the number of backlogged stations at any particular time as $n$.

# Backlogged Stations

- A station is considered backlogged between the time that it became aware that its previous transmission failed and the time of a successful transmission.
- Once a station is aware that a transmission was unsuccessful it waits for $T_i$ time units to retransmit where the distribution over these waits is given by

$$p\left(T_i\right) = \mu \exp\left(-\mu T_i\right).$$

# Backlogged Stations (continued)

- If $n$ nodes are backlogged then the arrivals from backlogged nodes is a Poisson process with rate $n\mu$.
- The overall 'arrival process' for the system is Poisson with a rate $G(n) = \lambda + n\mu$.

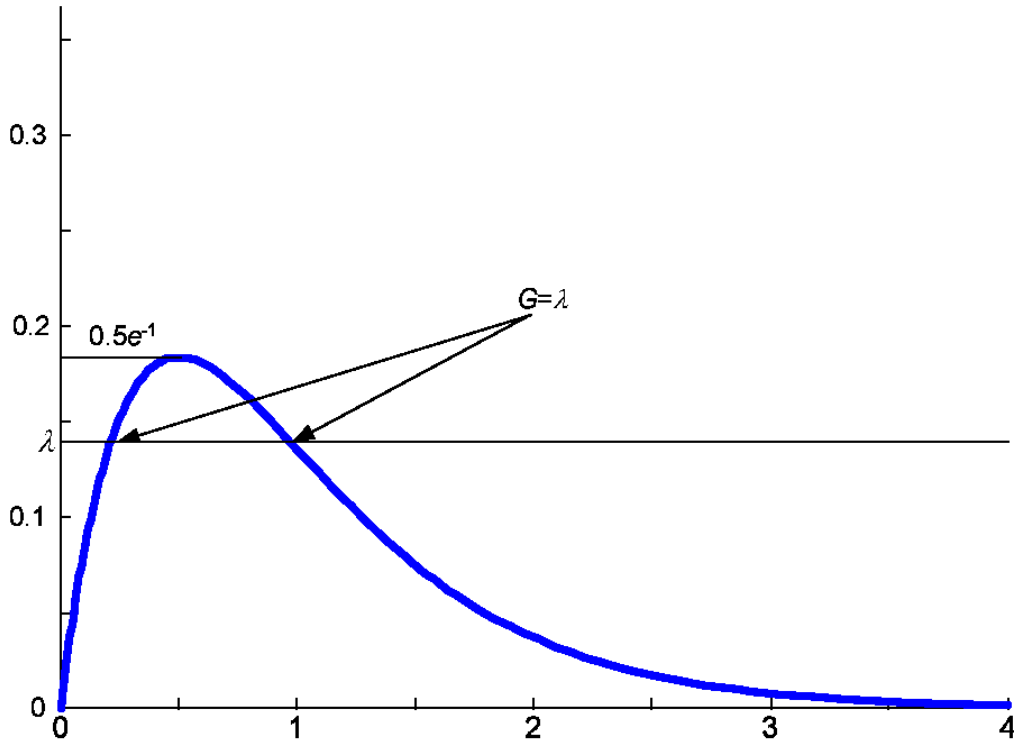
# Pure ALOHA

- The probability that there are no arrivals in an interval of 2 time units for a Poisson process with rate $G(n)$ is given by

$$p(k = 0|G(n), 2) = \exp(-2G(n)).$$

- Therefore this is the probability that there is no collision between the packet of interest and any other packets in the network.
- It is therefore the probability of a successful transmission.

# Throughput




pureALOHAThroughput

# Throughput (continued)

- The rate of attempted transmissions is $G(n)$ so the rate of successful transmissions is given by

$$G'(n) = G(n) \exp(-2G(n)),$$

- This is known as the throughput.
- It is the number of packets that successfully reach their destination. The throughput is plotted as a function of $G$ .
-

# Maximum Throughput

- The curve goes up to a maximum point and drops down thereafter.
- We can find the maximum of the throughput by looking for points where the gradient of the throughput is zero.
- Taking the gradient of $G'$ with respect to $G$ gives

$$\frac{dG'}{dG} = \exp(-2G) - 2G\exp(-2G).$$

# Stationary Points

- To find the stationary points we set this equation to zero and solve for $G$. The equation becomes

$$\exp(-2G) - 2G\exp(-2G) = 0$$

$$(1 - 2G)\exp(-2G) = 0$$

- So either $\exp(-2G) = 0$, which in turn implies that $G \to \infty$ or

# Stationary Points (continued)

- $1 - 2G = 0$ which implies $G = \frac{1}{2}$.
- The first solution is a minimum (and the throughput is zero) and the second solution is a maximum.
- The maximum is the maximum possible throughput for the system and is given when $G = \frac{1}{2}$.

# Maximum Throughput

- Substituting into the equation for the throughput we find that the maximum is $\frac{1}{2}\exp(-1) = \frac{1}{2e} \approx 0.184$. This maximum throughput shows that at best the channel is in use only about 18% of the time. [2]
- The system will be in equilibrium if the throughput equals $\lambda$.

[2] Note that since the service time is fixed at $X_n = 1$ for all $n$ then the utilisation for this system is given by $\rho = \lambda \bar{X} = \lambda$.

# Analysis of Backlog Delay

- Recall our assumption that a backlogged node waits for a time, $T_i$, before attempting to retransmit,

$$p\left(T_i\right) = \mu \exp\left(-\mu T_i\right).$$

- Given this assumption, can we compute the delay associated with this backlogged node?
- Since the node may experience several collisions, the wait associated with the backlog will involve a sum over the random variable $T_i$. It turns out to be an *Erlang distribution*.

# Sums of Exponential Variables and the Erlang Distribution

- If a random variable, $x$, is exponentially distributed with rate parameter $b$. The sum of $a$ random variables, each of which is exponentially distributed with rate parameter $b$ is distributed as an *Erlang distribution.*
- Consider variables a set of $a$ variables$(x_1 \dots x_a)$ ,

$$p\left(x_i|b\right) = b\exp\left(-bx_i\right).$$

# Sums of Exponential Variables and the Erlang Distribution (continued)

- We define the sum of the values to be, $z = \sum_{i=1}^{a} x_i$.
- The distribution governing $z$ is then an *Erlang distribution*, given by

$$p(z|a,b) = \frac{b^a}{(a-1)!} z^{a-1} \exp(-bz).$$

# Notes on Erlang

- Note that this distribution is a generalisation of the exponential distribution because the exponential is the special case where $a = 1$.
- This clearly should be the case because if $a = 1$ then the 'sum' of that variable will simply be the variable itself.

# Erlang Distribution Mean and Variance

- The mean of the Erlang distribution is given by

$$\langle z \rangle_{p(z|a,b)} = \frac{a}{b}$$

and the variance is

$$\mathrm{var}\,(z) = \frac{a}{b^2}.$$

# Delay Associated with Backlogs

- If the node retransmits $k$ times then the distribution over the total time required, $W = \sum_{i=1}^{k} T_i$, comes from the sum of $k$ independent exponentially distributed variables.
- We know that the resulting variable $X$ is Erlang distributed with parameters $\mu$ and $k$,

$$p\left(W|k,\mu\right) = \frac{\mu^k}{(a-1)!} W^{k-1} \exp\left(-\mu W\right).$$

- The probability of transmitting after $k$ attempts is given by

$$p\left(k\right) = q\left(1-q\right)^{k-1}.$$

where $q$ is the probability of a successful transmission. For pure ALOHA $q = \exp\left(-2G\right)$. [3]

[3] Compare this with the ARQ system we analysed in the chapter on $M/G/1$ queues. There we were considering errors (unsuccessful transmissions) so $p = (1-q)$.

# Distribution for $W$

- From the product rule and the sum rule we know that the probability distribution for $W$ is given by

$$p(W|\mu) = \sum_{k=1}^{\infty} p(W|k,\mu)\, p(k)\,,$$

$$p(W|\mu) = \sum_{k=1}^{\infty} q(1-q)^{k-1} \frac{\mu^k}{(k-1)!} W^{k-1} \exp(-\mu W)\,,$$

substituting $\dot{k} = k - 1$

$$p(W|\mu) = \sum_{k=0}^{\infty} q(1-q)^{\dot{k}} \frac{\mu^{\dot{k}+1}}{\dot{k}!} X^{\dot{k}} \exp(-\mu W)\,,$$

extracting the $q$, the exponential term and one of the $\mu$s from the product we rewrite as,

# Optimum value of $\mu$

- For a given value of $\lambda$ and $n$ there is a optimal value for $\mu$.
- This value of $\mu$ can be found by seeking the maximum rate, this is done by differentiating with respect to $\mu$

$$\frac{d\mu'}{d\mu} = \exp\left(-2\left(\lambda + n\mu\right)\right) - 2n\mu \exp\left(-2\left(\lambda + n\mu\right)\right)$$

and setting the result to zero to find the stationary points,

$$\exp\left(-2\left(\lambda + n\mu\right)\right) - 2n\mu \exp\left(-2\left(\lambda + n\mu\right)\right) = 0,$$

$$\left(1 - 2n\mu\right) \exp\left(-2\left(\lambda + n\mu\right)\right) = 0.$$

# Solutions for $\mu$

- There are two solutions to this equation. The first is $\exp\left(-2\left(\lambda + n\mu\right)\right) = 0$, meaning that $\lambda + n\mu \to \infty$. This solution is a minimiser . The other solution is given when

$$0 = 1 - 2n\mu$$

or

$$\mu = \frac{1}{2n},$$

this turns out to be a maximum of (38).

# Notes on Optimum

- By maximising the 'rate' parameter $\mu'$ we minimise the average delay associated with the backlog, which is given by $\bar{W} = \frac{1}{\mu'}$.
- By setting $\mu = \frac{1}{2n}$ we find a minimum for the average delay:

$$\bar{W}_{\min} = 2n \exp(2\lambda + 1)$$

.

# Notes on Optimum (continued)

- In practice it may be difficult to set the rate $\mu$ to something that is dependent on the number of backlogged nodes.
- This is because the number of backlogged nodes can only be known by nodes communicating their status over the network, we have not allowed for such a mechanism in our analysis.
- Note that the minimum average delay goes up linearly with the number of backlogged nodes and *exponentially* with the rate of arrivals for the system, $\lambda$.

# Slotted ALOHA

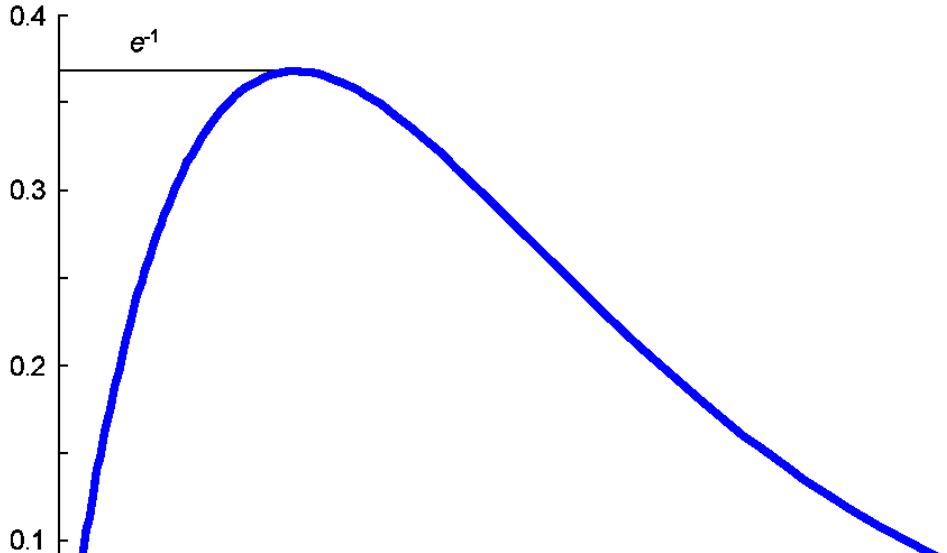
- A simple improvement to the ALOHA system comes from constraining when packets can be sent.
- Instead of allowing stations to transmit immediately they receive a packet each station waits for the beginning of a 'slot' to start transmitting.
- These slots are synchronised across the entire network.
- In this system a collision occurs if another station attempts to transmit in the same slot.
- This happens if another station received a packet for transmission in the preceding slot.
- If each slot is one time unit in length the probability of this event is given by

$$p\left(k=0|G\left(n\right),2\right)=\exp\left(-G\left(n\right)\right).$$

# Slotted ALOHA

- The probability is smaller than for pure ALOHA because the 'vulnerable period' for a collision is smaller (half the length of before).
- A collision now only occurs if another station attempts to transmit in the same slot. As a result the rate of successful transmissions is given by

$$G'(n) = G(n)\exp(-G(n)),$$

- 

# Smaller Vulnerable period

- The smaller vulnerable period leads to a greater maximum throughput which is now found at $G = 1$. The maximum throughput is $\frac{1}{e}$. Slotted ALOHA therefore can be twice as efficient as pure ALOHA.

# References


- 

# Thanks!

- company: Trent AI
- book: The Atomic Human
- twitter: @lawrennd
- The Atomic Human
- newspaper: Guardian Profile Page
- blog: http://inverseprobability.com

# Further Reading

- Multiaccess, ALOHA, and slotted ALOHA of Bertsekas and Gallager (1992)

# References

Bertsekas, D.P., Gallager, R.G., 1992. Data networks, Second. ed. Prentice-Hall, Upper Saddle River, NJ.